MAPPING RESEARCH ON INDUSTRY 4.0 TREND IN HEALTHCARE: A TOPIC MODELLING-BASED LITERATURE REVIEW

Abstract - Industry 4.0 has managed to revolutionize several industries by implementation of technologies like Internet of Things, Big Data Analytics, blockchain, Cloud Computing, and Artificial Intelligence. The healthcare and medical domain also benefit from this revolution. This paper aims to provide a comprehensive analysis of the literature on Industry 4.0 applications in the healthcare industry. The topic-modelling based literature review helps identify key topics of interest using the probabilistic topic model Latent Dirichlet Allocation (LDA). Five topics were identified and analysed to understand the pertinent technologies for the medical field.

Keywords: LDA, data mining, Industry 4.0, Internet of Things

I. Introduction

The Fourth Industrial Revolution, also referred to as Industry 4.0 aims to achieve innovation by integrating intelligent digital technologies into manufacturing and other industrial processes. Popular technologies in Industry 4.0 include the Internet of Things (IoT), big data analytics (BDA), blockchain, robotics and Artificial Intelligence (AI). The transformative impact of these technologies, which has reshaped numerous industrial sectors, holds significant promise for revolutionizing the healthcare landscape as well.

Healthcare 4.0 is a branch of Industry 4.0 that aims to revolutionize the healthcare industry by making it patient focused and efficient. It serves to improve the pre-medical diagnosis phase which helps with taking preventative measures for the patients.[62]. The technologies that exist under the Industry 4.0 umbrella can offer customised medical solutions pertaining to each patients need. The large scale of patient information will also benefit from Big Data solutions such as cloud storage [6]. The actual application of these technologies is largely dependent on how well the company and staff integrate it with the existing infrastructure [3]. Hence, there is a need for extensive analysis of how different Industry 4.0 technologies can benefit different medical sectors.

Amidst the abundance of information within the realm of Healthcare 4.0, employing text mining techniques becomes imperative for effectively sifting through the vast datasets and extracting actionable insights. Text mining can efficiently aid in discerning pertinent topics, thereby facilitating streamlined research processes, as highlighted by a study conducted by [5] which emphasized the potential of information extraction and text mining in generating diverse knowledge patterns. Topic modelling is a technique to discover the underlying themes present in a text corpus. Humans perceive

documents as a cluster of topics with different set of words signifying a relation to a certain topic. Topic modelling uses algorithms to mimic this concept. It provides a probabilistic framework using generative models. Latent semantic analysis, Non-Negative Matrix Factorization, Probabilistic Latent semantic analysis, and Latent Dirichlet Allocation are the most common topic modelling techniques that have contributed to every sector of text analysis in many areas [12].

II. Background

A. Industry 4.0 technologies

In this section, we provide a succinct overview of the diverse Industry 4.0 technologies utilized within the healthcare sector.

i. Internet of Things (IoT):

IOT comprises of physical objects and devices that enable connectivity and data exchange through embedded sensors and software. Data collection for diagnosis purposes can massively benefit from use of such devices [88].

ii. Artificial Intelligence (AI):

Artificial intelligence consists of computer systems that perform tasks by simulating human intelligence processes. Data prediction be it in diagnosis or planning of medical supply, can be easily simplified using technologies like machine learning [74].

iii. Big Data Analytics:

Big data analytics focuses on analyse large volumes of structured and unstructured data to uncover any existing trends and patterns. Genomics, clinical trials and personalized medication can especially benefit from such analytics and speed up the process for providing care to patients.

iv. Robotics and Automation:

Robotic systems and automated processes are developed to perform tasks traditionally carried out by humans and improve efficiency..

v. Augmented Reality (AR) and Virtual Reality (VR):

AR overlays digital content onto the real world, enhancing the user's perception, while VR immerses users in a completely virtual environment.

vi. Blockchain:

Blockchain is a decentralized, distributed ledger technology that enables secure and transparent recording of transactions across multiple parties. By removing the need for a central party, blockchain improves transaction speeds, eradicates the single point of failure and reduces transaction costs [44].

vii. 3D Printing:

It involves the creation of three-dimensional objects by layering materials based on digital models. Covid 19 emphasized the importance of such technology. 3D printing were used to produce PPE kits and parts for ventilators after the failure of traditional supply chains [53].

viii. Cloud Computing:

Cloud computing comprises of a shared pool of computing resources, including servers, storage, and applications that are delivered over the internet on demand. Scalability and flexibility are the two main selling point for this technology. This is especially beneficial to the healthcare industry which often faces the challenge of being understaffed and overworked.

B. Latent Dirichlet Allocation

LDA is a three-level hierarchical Bayesian model for collections of discrete data such as text corpora [13]. The LDA model is represented as a probabilistic graphical model in Figure 1.

LDA assumes the following generative process for each document w in a corpus D:

1. Choose N ~ Poisson(ξ).

2. Choose θ ~ Dir(α).

3. For each of the N words w_n:

   (a) Choose a topic z_n ~ Multinomial(θ).

   (b) Choose a word w_n from p (w_n |z_n, β), a multinomial probability conditioned on the topic z_n.

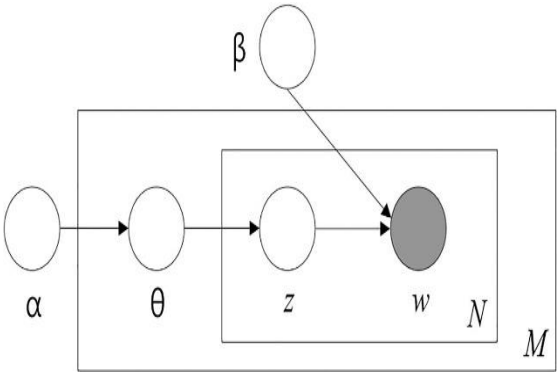


Figure 1: Model representation of LDA

III. Research method

This section gives the theoretical background, data analysis, and details about the topic model implemented. The aim is to create a topic model framework that can be generalized and applied for exploratory literature review.

A. Data sources

The data collection process involved methodically searching through various academic databases like IEEE Explore, Springer, Elsevier, and other sources. Specific keywords were utilized for this search – "Healthcare 4.0," "Industry 4.0 healthcare" and "Healthcare research paper." Articles which were not directly pertaining to healthcare were excluded. Studies that are not written in the English language or any regional native language are also excluded from the study.

B. Text mining

One hundred research papers constituted the primary dataset, from which text was extracted for analysis. The extracted text excluded generalized information and references sections from each research paper. Each entry in the dataset corresponds to a distinct research paper, facilitating focused analysis on individual research contributions.

C. Data Preprocessing

The sentences in each of the research paper are further pre-processed. The preprocessing is done to prepare the input for the LDA model which cannot directly take textual input. This preprocessing stage involves multiple steps. All words were converted to lower case followed by the removal of punctuation and any digits. A specialized list of stopwords was created which contained common non-contributing words related to the healthcare industry. These words were removed from the text along with the standard list of English stopwords. Lemmatization was then performed on the text.

D. Keyword Identification

To further reduce the amount of non- contributing words and to highlight the important words occurring in each document, keywords were identified for each record. A KeyBERT model was utilized for this step and 70 key phrases were extracted for each document with a ngram

range of [1,3]. BERT embeddings are used by this model to extract key phrases most relevant to the document.

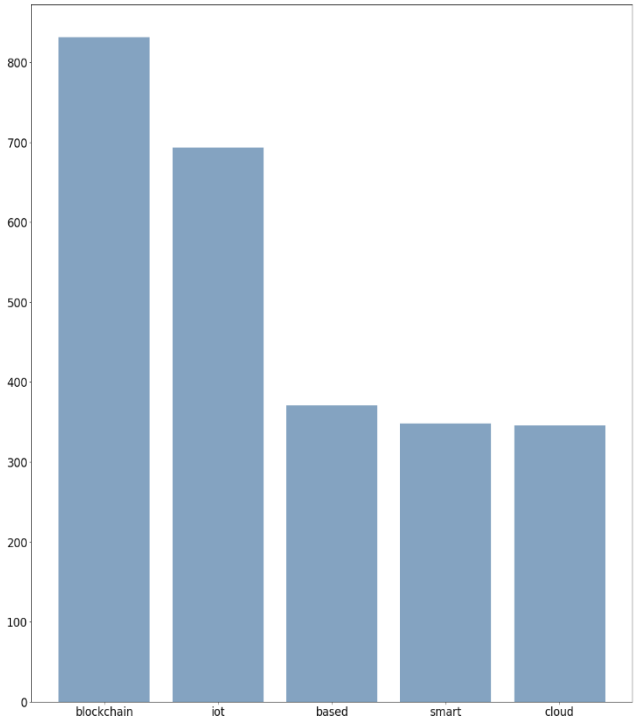


Figure 2: Top five words with most occurrence in the corpus after keyword extraction

E. LDA model

The processed text from each document in then converted to a matrix of TF-IDF features. This matric is used by the LDA model as an input. The total number of topics to be identified also needs to be provided as a parameter to the LDA model. The optimal number of topics for this set of documents was decided to be five after testing out different topic numbers.

The model was run for 200 iterations and top 6 words were identified for each topic. The documents were then analysed based on the topics identified. The topic with maximum probability was assigned as the major theme for that document.

IV. Results

A. Descriptive analysis

Recent years have seen a significant increase in publications on healthcare 4.0, indicating a growing academic interest in exploring technological advancements for improving medical services. Healthcare 4.0 emerged in the wake of Industry 4.0, gaining traction in the mid-2010s.

Notably, the year 2023 witnessed a significant surge in publication activity, with most articles in this review being published during this period. This surge reflects the maturation of previous ambitious concepts into tangible and practical solutions, facilitated by substantial technological advancements over the preceding five years. Figure 3 denotes the distribution of the papers over the years.

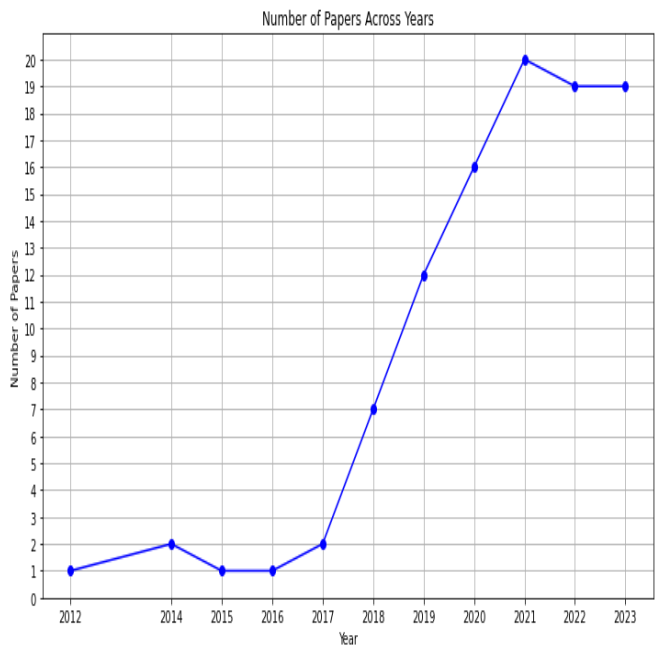


Figure 3: Distribution of papers across the years

B. Thematic analysis

i. Topic 1 – Medical analytics

Insufficient information often leaves patients and clinicians unable to discern which strategies offer benefits and which may be detrimental, underscoring the need for advanced analytics and personalized predictions to optimize decision-making in healthcare [54].

[29] outlines six use cases for leveraging medical analytics, particularly predictive systems, in healthcare. These include managing high-cost patients, reducing readmissions, improving triage processes, identifying patient decompensation, preventing adverse events, and optimizing treatment for diseases affecting multiple organ systems.

ii. Topic 2 –Cloud computing and IOT

IOT and cloud computing are the most prevalent technologies present in the medical sphere. By shifting data to the cloud, hospitals, pharmacies, and insurance companies may conveniently communicate patient medical records, prescription information and test results which can be accessed by authorised organisations from anywhere [50].

Data of such magnitude comes with its own set of problems. Data security is a major issue with respect to the data that IOT devices keep generating.[57] Hence authentication and access control needs to be properly setup in order to protect the sensitive patient data being uploaded to the cloud.

iii. Topic 3 – Blockchain

Security is a prominent issue when it comes to healthcare. Financial information, medical records and important drug information can all be tampered with when it comes to digital methods of storing data. Integration of blockchain with traditional healthcare, particularly in the public sector can help increase transparency and make public healthcare more ethical and efficient [61]. Blockchain-based cloud solutions like MeDShare[49] aim to reduce the risk of tampering in cloud-based information sharing.

Another compelling application for blockchain-based solutions is in the realm of data exchange. Typically, data exchange between diverse e-health entities poses significant challenges in terms of security, privacy, and network loads, particularly in large-scale healthcare systems. Existing solutions often suffer from poor scalability, high computational costs, and slow response times. [90] proposes a framework that addresses these challenges by integrating edge computing and blockchain technologies

Analysis of our text corpus under this topic reveals that despite the identification of several use cases for blockchain in healthcare, most research is in the prototype stage [44].

iv. Topic 4 –Robotics and Sensors

Wearable sensors can measure physiological biomarkers which enables personalization of treatment. This detailed information can be uploaded to the cloud, updated in real time, and accessed through mobile applications, thus developing a smart healthcare system.

Robotics can provide a non-invasive way of doing medical tasks. Researchers from the University of Bristol are studying how swarms of nanorobots could detect cancer cells and carry out non-invasive surgical interventions at cellular level in patients with tumours [41].

v. Topic 5 – Artificial Intelligence

Integrating artificial intelligence into telemedicine platforms enhances remote diagnosis and treatment capabilities while ensuring efficient patient care delivery and data security in the public healthcare system [74].

However, the integration of artificial intelligence needs to be done with several precautions and systems of check in place. An error in the AI system in the healthcare field

could have serious implications and lead to loss of life due to wrong diagnosis in a worst-case scenario. AI's purported ability to mitigate human error is countered by apprehensions surrounding discrimination and stigmatization, particularly when algorithms are not tailored to diverse populations, potentially perpetuating biases [76]. Moreover, ongoing concerns about privacy breaches and data manipulation underscore the need for vigilant monitoring and safeguards in AI systems.

In the realm of healthcare education, mobile learning has become indispensable, exemplified by the proliferation of m-learning applications tailored to medical curricula. For instance, medical students can access interactive anatomy modules and virtual patient simulations directly from their smartphones, facilitating immersive learning experiences anytime, anywhere.

Moreover, healthcare professionals' benefit from mobile systems that provide access to evidence-based guidelines, medical literature, and diagnostic tools, enabling continuous learning and informed decision-making at the point of care. As mobile technologies continue to evolve, the integration of artificial intelligence solutions holds promise for personalized learning pathways and real-time clinical decision support, revolutionizing the landscape of healthcare education and practice [18].

Topic number	Topic words
1	['analytics' 'big' 'predict' 'knowledge' 'advanced' 'potential']
2	['iot' 'manufacturing' 'smart' 'industrial' 'cloud' 'computing']
3	['blockchain' 'based' 'privacy' 'learning' 'network' 'algorithm']
4	['robot' 'sensor' 'wearable' 'machine' 'learning' 'monitoring']
5	['artificial' 'intelligence' 'mobile' 'learning' 'machine' 'future']

Table 1: Topic words for each identified topic

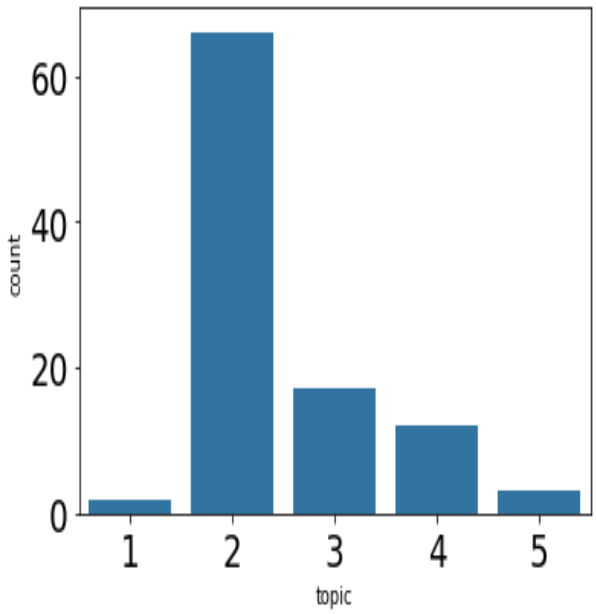


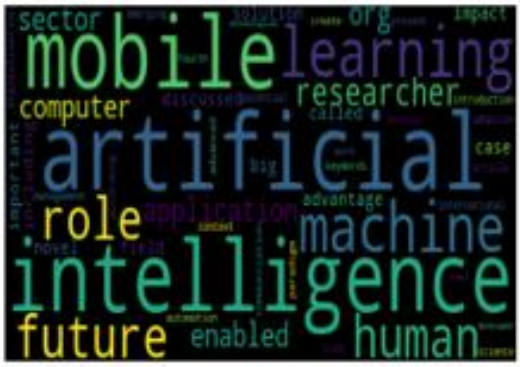
Figure 4: Distribution of research articles over different topics.

TOPIC 1

TOPIC 2

TOPIC 3

TOPIC 4



TOPIC 5

Figure 5: Word clouds for each of the six topics in the corpora

V. Future scope

More articles could be included in the future to enhance the scope of this study and find more technologies pertinent to healthcare. More region focused research can also be conducted to ensure that healthcare operations are optimized in a way that considers the differences among patients across several geographical locations.

VI. Conclusion

The study analysed research articles focused on healthcare 4.0 to identify key technologies using text mining and topic modelling through LDA. This analysis can help early-career researchers in the medical field identify the technology on which they wish to focus and familiarise themselves with the appropriate literature for the same.

References

1. R. Chinorický, J. Kurotová, and P. Janošková, "Measuring the impact of digital technologies on transport industry–macroeconomic perspective," Transportation Research Procedia, vol. 55, pp. 434-441, 2021, doi: 10.1016/j.trpro.2021.07.092.
2. R. Pershina, B. Soppe, and T. Thune, "Bridging analog and digital expertise: Cross-domain collaboration and boundary-spanning tools in the creation of digital innovation," Research Policy, vol. 48, 2019, doi: 10.1016/j.respol.2019.103819.
3. L. Barnewold and B. G. Lottermoser, "Identification of digital technologies and digitalisation trends in the mining industry," International Journal of Mining Science and Technology, vol. 30, no. 6, 2020.
4. T. Matsumoto, W. Sunayama, Y. Hatanaka, and K. Ogohara, "Data Analysis Support by Combining Data Mining and Text Mining," in 2017 6th IIAI International Congress on Advanced Applied Informatics (IIAI-AAI), Hamamatsu, Japan, 2017, pp. 313-318.
5. S. Salloum, M. Al-Emran, A. Monem, and K. Shaalan, "Using Text Mining Techniques for Extracting Information from Research Articles," 2018.
6. K. Lu and H. Liao, "A survey of group decision making methods in Healthcare Industry 4.0: bibliometrics, applications, and directions," Appl Intell, vol. 52, pp. 13689–13713, 2022.
7. Paul, M. Riffat, A. Yasir, M. N. Mahim, B. Y. Sharnali, I. T. Naheen, A. Rahman, and A. Kulkarni, "Industry 4.0 Applications for Medical/Healthcare Services," J. Sens. Actuator Netw., vol. 10, no. 3, p. 43, 2021. [Online]. Available: https://doi.org/10.3390/jsan10030043.
8. A. Haleem, M. Javaid, and R. Vaishya, "Industry 4.0 and its applications in orthopaedics," J. Clin. Orthop. Trauma, vol. 10, no. 3, pp. 615–616, 2019.
9. X. Wei and W. Croft, "LDA-based document models for Ad-hoc retrieval," in Proceedings of the 29th Annual International ACM SIGIR, 2006, pp. 178-185, doi: 10.1145/1148170.1148204.
10. Z. Tong and H. Zhang, "A Text Mining Research Based on LDA Topic Modelling," Computer Science & Information Technology, vol. 6, pp. 201-210, 2016, doi: 10.5121/csit.2016.60616.
11. E. Ikonomakis, S. Kotsiantis, and V. Tampakas, "Text Classification Using Machine Learning Techniques," WSEAS Transactions on Computers, vol. 4, pp. 966-974, 2005.
12. P. Kherwa and P. Bansal, "Topic Modeling: A Comprehensive Review," ICST Transactions on Scalable Information Systems, vol. 7, p. 159623, 2018.
13. D. Blei, A. Ng, and M. Jordan, "Latent Dirichlet Allocation," Journal of Machine Learning Research, vol. 3, pp. 993-1022, 2003.
14. Moosavi, J., Bakhshi, J., and Martek, I., "The application of industry 4.0 technologies in pandemic

management: Literature review and case study," Healthcare Analytics, vol. 1, p. 100008, 2021, doi: 10.1016/j.health.2021.100008.

15. P. Pace, G. Aloi, R. Gravina, G. Caliciuri, G. Fortino, and A. Liotta, "An Edge-Based Architecture to Support Efficient Applications for Healthcare Industry 4.0," in IEEE Transactions on Industrial Informatics, vol. 15, no. 1, pp. 481-489, Jan. 2019, doi: 10.1109/TII.2018.2843169.
16. A. Sarfraz, Z. Sarfraz, M. Sarfraz, A. Razzack, S. Bano, S. Singh Makkar, S. Thevuthasan, T. Paul, M. K. Sana, N. Azeem, M. Felix, and I. Chérrez-Ojeda, "Industry 4.0 Technologies for the Manufacturing and Distribution of COVID-19 Vaccines," *Journal of Primary Care & Community Health*, vol. 13, 2022, doi: 10.1177/21501319211068638.
17. G. Aceto, V. Persico, and A. Pescapè, "Industry 4.0 and Health: Internet of Things, Big Data, and Cloud Computing for Healthcare 4.0," *Journal of Industrial Information Integration*, vol. 18, p. 100129, 2020, doi: 10.1016/j.jii.2020.100129.
18. Mateus-Coelho, Nuno, & Avila, Paulo. (2021). "Application of the Industry 4.0 Technologies to Mobile Learning and Health Education Apps." *FME Transactions*, 49, 877. doi: 10.5937/fme2104876M.
19. S. Khan, I. Saied, T. Ratnarajah, and T. Arslan, "Evaluation of Unobtrusive Microwave Sensors in Healthcare 4.0-Toward the Creation of Digital-Twin Model," *Sensors*, vol. 22, p. 8519, 2022, doi: 10.3390/s22218519.
20. L. Lhotska, "Application of Industry 4.0 Concept to Health Care," *Studies in health technology and informatics*, vol. 273, pp. 23-37, 2020, doi: 10.3233/SHTI200613.
21. B. Alhayani, ·, S. Kwekha-Rashid, ·, ·, B. Mahajan, H. Ilhan, N. Uke, A. Alkhayyat, H. J. Mohammed, J. Mohammed, and H. Mahajan, "5G standards for the Industry 4.0 enabled communication systems using artificial intelligence: perspective of smart healthcare system of Things," *Applied Nanoscience*, vol. 13, no. 3, 2022, doi: 10.1007/s13204-021-02152-4.
22. M. Javaid, A. Haleem, R. Vaishya, S. Bahl, R. Suman, and A. Vaish, "Industry 4.0 technologies and their applications in fighting COVID-19 pandemic," *Diabetes and Metabolic Syndrome Clinical Research and Reviews*, vol. 14, 2020, doi: 10.1016/j.dsx.2020.04.032.
23. Paranitharan, K.P., Ebenezer, G., Venkataraman, Balaji, Khan, M., & Thangavelu, Ramesh. (2022). "Application of industry 4.0 technology in containing Covid-19 spread and its challenges." *Materials Today: Proceedings*, 68. doi: 10.1016/j.matpr.2022.06.009.
24. Ahmad, Muhammad, Sadiq, Saima, Eshmawi, Ala, Alluhaidan, Ala, Umer, Muhammad, Ullah, Dr. Saleem, & Nappi, Michele. (2022). "Industry 4.0 technologies and their applications in fighting COVID-19 pandemic using deep learning techniques." *Computers in Biology and Medicine*, 145, 105418. doi: 10.1016/j.compbiomed.2022.105418.
25. A. Chauhan, S. Jakhar, and C. Chauhan, "The interplay of circular economy with industry 4.0 enabled smart city drivers of healthcare waste disposal," *Journal of Cleaner Production*, vol. 279, p. 123854, 2020, doi: 10.1016/j.jclepro.2020.123854.
26. C. Chute and T. French, "Introducing Care 4.0: An Integrated Care Paradigm Built on Industry 4.0 Capabilities," Int. J. Environ. Res. Public Health, vol. 16, no. 12, pp. 2247, Jun. 2019, doi: 10.3390/ijerph16122247.
27. M. D. Venkata, S. Lingamgunta, and K. Murali, "Health Care Automation in Compliance to Industry 4.0 Standards: A Case Study of Liver Disease Prediction," *Journal of Scientific & Industrial Research*, vol. 82, no. 2, pp. 263-268, Feb. 2023, doi: 10.56042/jsir.v82i2.70215.

28. E. Onyema, Dr. K. Lilhore, P. Saurabh, S. Dalal, A. Nwaeze, A. Chijindu, L. Ndufeiya-Kumasi, and S. Simaiya, "Evaluation of IoT-Enabled hybrid model for genome sequence analysis of patients in healthcare 4.0," *Measurement: Sensors*, vol. 26, p. 100679, 2023, doi: 10.1016/j.measen.2023.100679.
29. D. Bates, S. Saria, L. Ohno-Machado, and A. Shah, "Big Data in Health Care: Using Analytics to Identify and Manage High-Risk and High-Cost Patients," *Health affairs (Project Hope)*, vol. 33, pp. 1123-1131, 2014, doi: 10.1377/hlthaff.2014.0041.
30. V. Kashyap, A. Kumar, A. Kumar, and Y.-C. Hu, "A Systematic Survey on Fog and IoT Driven Healthcare: Open Challenges and Research Issues," *Electronics*, vol. 11, no. 17, p. 2668, 2022, doi: 10.3390/electronics11172668.
31. J. Li and P. Carayon, "Health Care 4.0: A Vision for Smart and Connected Health Care," *IISE Transactions on Healthcare Systems Engineering*, vol. 11, pp. 1-14, 2021, doi: 10.1080/24725579.2021.1884627.
32. A. Gupta and A. Singh, "Healthcare 4.0: Recent advancements and futuristic research directions," Wireless Personal Communications, vol. 129, no. 2, pp. 933-952, 2023, doi: 10.1007/s11277-022-10164-8.
33. A. Haleem, M. Javaid, R. Singh, and R. Suman, "Medical 4.0 technologies for healthcare: Features, capabilities, and applications," Internet of Things and Cyber-Physical Systems, vol. 2, 2022, doi: 10.1016/j.iotcps.2022.04.001.
34. K. Kotzias, F. A. Bukhsh, J. J. Arachchige, M. Daneva, and A. Abhishta, "Industry 4.0 and healthcare: Context, applications, benefits and challenges," IET Software, vol. 17, no. 3, pp. 195-248, 2023.
35. S. Tanwar, K. Parekh, and R. Evans, "Blockchain-based electronic healthcare record system for healthcare 4.0 applications," *Journal of Information Security and Applications*, vol. 50, p. 102407, 2020, doi: 10.1016/j.jisa.2019.102407.
36. M. U. Rehman, A. E. Andargoli, and H. Pousti, "Healthcare 4.0: Trends, Challenges and Benefits," in Proceedings of the Australasian Conference on Information Systems, 2019, pp. 556, Perth, Western Australia
37. G. L. Tortorella, F. S. Fogliatto, T. A. Saurin, L. M. Tonetto, and D. McFarlane, "Contributions of Healthcare 4.0 digital applications to the resilience of healthcare organizations during the COVID-19 outbreak," Technovation, vol. 111, pp. 102379, Mar. 2022, doi: 10.1016/j.technovation.2021.102379.
38. J. Al-Jaroodi, N. Mohamed, N. Kesserwan, and I. Jawhar, "Healthcare 4.0 – Managing a Holistic Transformation," pp. 1-8, 2022, doi: 10.1109/SysCon53536.2022.9773863.
39. M. M. Ahsan and Z. Siddique, "Machine learning-based heart disease diagnosis: A systematic literature review," Artificial Intelligence in Medicine, vol. 128, pp. 102289, Jun. 2022, doi: 10.1016/j.artmed.2022.102289.
40. G. Yang, C. Li, and K. E. Marstein, "A blockchain-based architecture for securing electronic health record systems," Concurrency and Computation: Practice and Experience, vol. 33, p. e5479, 2021.
41. C. Soler, "Health 4.0 Oriented to Non-surgical Treatment," in *Proceedings of the International Conference on Social Robotics (ICSR)*, 2018, pp. 518-527. doi: 10.1007/978-3-319-70836-2_42.
42. R. Rudrapati, "Using industrial 4.0 technologies to combat the COVID-19 pandemic," *Annals of Medicine and Surgery*, vol. 78, p. 103811, 2022, doi: 10.1016/j.amsu.2022.103811.
43. M. Ahmed, M. Kashem, M. Rahman, and S. Khatun, "Review and Analysis of Risk Factor of Maternal Health in Remote Area Using the Internet of Things (IoT)," 2020, doi: 10.1007/978-981-15-2317-5_30.
44. C. C. Agbo, Q. H. Mahmoud, and J. M. Eklund, "Blockchain Technology in Healthcare: A

Systematic Review," *Healthcare (Basel)*, vol. 7, no. 2, p. 56, Apr. 2019, doi: 10.3390/healthcare7020056.

45. M. Hassanalieragh, A. Page, T. Soyata, G. Sharma, M. Aktas, G. Mateos, B. Kantarci, and S. Andreescu, "Health Monitoring and Management Using Internet-of-Things (IoT) Sensing with Cloud-Based Processing: Opportunities and Challenges," 2015, doi: 10.1109/SCC.2015.47.
46. U. Hariharan, R. Kotteswaran, A. T, and J. Janardhanan, "Smart Wearable Devices for Remote Patient Monitoring in Healthcare 4.0," 2021, doi: 10.1007/978-3-030-63937-2_7.
47. S. Chand and Y. Zhang, "Learning from machines to close the gap between funding and expenditure in the Australian National Disability Insurance Scheme," *Int. J. Inf. Manage. Data Insights*, vol. 2, no. 1, pp. 100077, 2022. doi: 10.1016/j.jjimei.2022.100077.
48. H. Asri, "Big Data and IoT for real-time miscarriage prediction: A clustering comparative study," Procedia Computer Science, vol. 191, pp. 200-206, 2021, doi: 10.1016/j.procs.2021.07.025.
49. Q. Xia, E. Sifah, K. Asamoah, J. Gao, X. Du, and M. Guizani, "MeDShare: Trust-less Medical Data Sharing Among Cloud Service Providers Via Blockchain," *IEEE Access*, pp. 1-1, 2017, doi: 10.1109/ACCESS.2017.2730843.
50. S. Ahuja, S. Mani, and J. Zambrano, "A Survey of the State of Cloud Computing in Healthcare," *Network and Communication Technologies*, vol. 1, pp. 12, 2012, doi: 10.5539/nct.v1n2p12.
51. M. Mahmud, M. S. Kaiser, M. M. Rahman, M. Rahman, A. Shabut, S. Al Mamun, and A. Hussain, "A Brain-Inspired Trust Management Model to Assure Security in a Cloud Based IoT Framework for Neuroscience Applications," *Cognitive Computation*, 2018, doi: 10.1007/s12559-018-9543-3.
52. J. Carvalho, Á. Rocha, J. B. de Vasconcelos, and A. Abreu, "A health data analytics maturity model for hospitals information systems," *International Journal of Information Management*, vol. 46, pp. 278-285, 2019, doi: 10.1016/j.ijinfomgt.2018.07.001.
53. L. Darwish, M. Farag, and M. T. El-Wakad, "Towards Reinforcing Healthcare 4.0: A Green Real-Time IIoT Scheduling and Nesting Architecture for COVID-19 Large-Scale 3D Printing Tasks," *IEEE Access*, vol. 8, pp. 213916-213927, 2020, doi: 10.1109/ACCESS.2020.3040544.
54. H. Krumholz, "Big Data And New Knowledge In Medicine: The Thinking, Training, And Tools Needed For A Learning Health System," *Health affairs (Project Hope)*, vol. 33, pp. 1163-1170, 2014, doi: 10.1377/hlthaff.2014.0053.
55. M. Stankovic, J. Gavrilovic, D. Jovanovic, M. Zivkovic, M. Antonijevic, N. Bacanin, and M. Stankovic, "Tuning Multi-Layer Perceptron by Hybridized Arithmetic Optimization Algorithm for Healthcare 4.0," *Procedia Computer Science*, vol. 215, pp. 51-60, 2022, doi: 10.1016/j.procs.2022.12.006.
56. A. M. AbdelMouty, A. Abdel-Monem, S. Aal, and M. Ismail, "Analysis the Role of the Internet of Things and Industry 4.0 in Healthcare Supply Chain Using Neutrosophic Sets," *Neutrosophic Systems with Applications*, vol. 4, pp. 33-42, 2023, doi: 10.61356/j.nswa.2023.15.
57. G. Manogaran, C. Thota, D. Lopez, and R. Sundarasekar, "Big Data Security Intelligence for Healthcare Industry 4.0," 2017, doi: 10.1007/978-3-319-50660-9_5.
58. S. Paul, M. Riffat, A. Yasir, M. N. Mahim, B. Y. Sharnali, I. T. Naheen, A. Rahman, and A. Kulkarni, "Industry 4.0 Applications for Medical/Healthcare Services," *J. Sens. Actuator Netw.*, vol. 10, no. 3, p. 43, 2021, doi: 10.3390/jsan10030043.
59. F. Abdullayeva, "Internet of Things-based healthcare system on patient demographic data in

Health 4.0," CAAI Transactions on Intelligence Technology, vol. 7, 2022, doi: 10.1049/cit2.12128.

60. V. Puri, A. Kataria, and V. Sharma, "Artificial intelligence-powered decentralized framework for Internet of Things in Healthcare 4.0," *Transactions on Emerging Telecommunications Technologies*, 2021, doi: 10.1002/ett.4245.
61. A. Kumar, R. Krishnamurthi, A. Nayyar, K. Sharma, V. Grover, and E. Hossain, "A Novel Smart Healthcare Design, Simulation, and Implementation Using Healthcare 4.0 Processes," *IEEE Access*, vol. 8, pp. 118433-118471, 2020, doi: 10.1109/ACCESS.2020.3004790.
62. S. Weerabahu, P. Samaranayake, R. Wickramarachchi, and T. Ilangakoon, "Adoption of Industry 4.0 and lean concepts in hospitals for healthcare operational performance improvement," *International Journal of Productivity and Performance Management*, pp. ahead-of-print, 2021, doi: 10.1108/IJPPM-12-2020-0654.
63. H.B. Mahajan, A.S. Rashid, A.A. Junnarkar, N. Uke, S.D. Deshpande, P.R. Futane, A. Alkhayyat, and B. Alhayani, "Integration of Healthcare 4.0 and blockchain into secure cloud-based electronic health records systems," *Appl Nanosci.*, vol. 13, no. 3, pp. 2329-2342, 2023, doi: 10.1007/s13204-021-02164-0.
64. H. Abrar, S. Hussain, J. Chaudhry, K. Saleem, M. Orgun, J. Al-Muhtadi, and C. Valli, "Risk Analysis of Cloud Sourcing in Healthcare and Public Health Industry," *IEEE Access*, pp. 1-1, 2018, doi: 10.1109/ACCESS.2018.2805919.
65. N. Uke, P. Pise, H. Mahajan, S. Harale, and S. Uke, "Healthcare 4.0 Enabled Lightweight Security Provisions for Medical Data Processing," *Turkish Journal of Computer and Mathematics Education (TURCOMAT)*, vol. 12, pp. 165-173, 2021, doi: 10.17762/turcomat.v12i11.5858.
66. M. Javaid and A. Haleem, "Impact of Industry 4.0 to create advancements in orthopaedics," *Journal of Clinical Orthopaedics and Trauma*, vol. 11, 2020, doi: 10.1016/j.jcot.2020.03.006.
67. E. Talhi, J.-C. Huet, V. Fortineau, and S. Lamouri, "A methodology for cloud manufacturing architecture in the context of Industry 4.0," *Bulletin of the Polish Academy of Sciences, Technical Sciences*, vol. 68, pp. 271-284, 2020. doi: 10.24425/bpasts.2020.131849.
68. S. Qahtan, K. Yatim, M.H. Osman, H. Zulzalil, M. Luqman, and M. Zakaria, "A Decision Cloud Ranking Approach Based on Privacy and Security in Blockchain E-Health Industry 4.0 Systems," *Journal of Techniques*, vol. 5, pp. 1-15, 2023, doi: 10.51173/jt.v5i4.1464.
69. I. Kaya and S. Demir, "Health 4.0: Blockchain Applications in Healthcare," *Journal of Investigations on Engineering & Technology* ,vol. 4, no. 2, pp. 69-76, Dec. 2021, doi: 10.2687-3052.
70. P. Bhattacharya, S. Tanwar, U. Bodkhe, S. Tyagi, and N. Kumar, "BinDaaS: Blockchain-Based Deep-Learning as-a-Service in Healthcare 4.0 Applications," IEEE/ACM Transactions on Networking, pp. 1-1, 2019, doi: 10.1109/TNSE.2019.2961932.
71. M. S. Islam, M. A. B. Ameedeen, M. A. Rahman, H. Ajra, and Z. B. Ismail, "Healthcare-Chain: Blockchain-Enabled Decentralized Trustworthy System in Healthcare Management Industry 4.0 with Cyber Safeguard," *Computers*, vol. 12, no. 2, p. 46, 2023, doi: 10.3390/computers12020046.
72. R. Gupta, A. Shukla, P. Mehta, P. Bhattacharya, S. Tanwar, S. Tyagi, and N. Kumar, "VAHAK: A Blockchain-based Outdoor Delivery Scheme using UAV for Healthcare 4.0 Services," in *Proceedings of IEEE INFOCOM Workshops*, 2020, pp. 370-375.
73. K. Mohit, H. Raj, N. Chaurasia, and S. S. Gill, "Blockchain inspired secure and reliable data exchange architecture for cyber-physical healthcare system 4.0," pp. 1-1, 2023, doi: 10.1016/j.iotcps.2023.05.006.

74. A. Kumar, N. Gupta, and P. Pujari, "Artificial Intelligence: Technology 4.0 as a solution for healthcare workers during COVID-19 pandemic Introduction -COVID-19 Pandemic," *Acta Universitatis Bohemiae Meridionalis*, vol. 24, pp. 1-10, 2021. doi: 10.32725/acta.2021.002.
75. P. Radanliev and D. De Roure, "Disease X vaccine production and supply chains: risk assessing healthcare systems operating with artificial intelligence and industry 4.0," *Health Technol (Berl)*, vol. 13, no. 1, pp. 11-15, 2023, doi: 10.1007/s12553-022-00722-2.
76. S. Mahomed, "Healthcare, artificial intelligence and the Fourth Industrial Revolution: Ethical, social and legal considerations," *South African Journal of Bioethics and Law*, vol. 11, p. 93, 2018, doi: 10.7196/SAJBL.2018.v11i2.00664.
77. S. Mahomed, "Healthcare, artificial intelligence and the Fourth Industrial Revolution: Ethical, social and legal considerations," South African Journal of Bioethics and Law, vol. 11, p. 93, 2018, doi: 10.7196/SAJBL.2018.v11i2.00664.
78. G. Yang et al., "Homecare Robotic Systems for Healthcare 4.0: Visions and Enabling Technologies," in *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 9, pp. 2535-2549, Sept. 2020, doi: 10.1109/JBHI.2020.2990529.
79. P. Dossou, L. Foreste, and E. Misumi, "Intelligent Support System for Healthcare Logistics 4.0 Optimization in the Covid Pandemic Context," *Journal of Software Engineering and Applications*, vol. 14, pp. 233-256, 2021, doi: 10.4236/jsea.2021.146014.
80. M. Fatima, N.U.K. Sherwani, S. Khan, and M.Z. Khan, "Assessing and predicting operation variables for doctors employing industry 4.0 in health care industry using an adaptive neuro-fuzzy inference system (ANFIS) approach," *Sustainable Operations and Computers*, vol. 3, pp. 286-295, 2022, doi: 10.1016/j.susoc.2022.05.005.
81. T. Ilangakoon, S. Weerabahu, and R. Wickramarachchi, "Combining Industry 4.0 with Lean Healthcare to Optimize Operational Performance of Sri Lankan Healthcare Industry," pp. 1-8, 2018, doi: 10.1109/POMS.2018.8629460.
82. E. D. Oña, J. M. Garcia-Haro, A. Jardón, and C. Balaguer, "Robotics in Health Care: Perspectives of Robot-Aided Interventions in Clinical Practice for Rehabilitation of Upper Limbs," *Applied Sciences*, 2019. [Online].
83. A. Soares, N. Piçarra, J.C. Giger, R. Oliveira, and P. Arriaga, "Ethics 4.0: Ethical Dilemmas in Healthcare Mediated by Social Robots," *Int J Soc Robot*, vol. 15, no. 5, pp. 807-823, 2023. [Online]. Available: https://doi.org/10.1007/s12369-023-00983-5
84. M. Pohl, "Robotic Systems in Healthcare with Particular Reference to Innovation in the 'Fourth Industrial Revolution'—An Ethical Challenge for Management," *Journal of International and Advanced Japanese Studies*, vol. 8, pp. 17–33, Feb. 2016. [Online].
85. P. Ajmera and V. Jain, "Modelling the barriers of Health 4.0–the fourth healthcare industrial revolution in India by TISM," *Oper Manag Res*, vol. 12, pp. 129-145, 2019, doi: 10.1007/s12063-019-00143-x.
86. T. Weerasinghe, K. Vidanagamachchi, and L. Nanayakkara, "Employee Competencies Development Framework for Industry 4.0 Adaptation in the Healthcare Sector," in *Proceedings of the IEEE*, 2020.
87. X. Larrucea, M. Moffie, S. Asaf, and I. Santamaria, "Towards a GDPR compliant way to secure European cross border Healthcare Industry 4.0," *Computer Standards & Interfaces*, vol. 69, p. 103408, 2019, doi: 10.1016/j.csi.2019.103408.
88. R. Jain, M. Gupta, A. Nayyar, and N. Sharma, "Adoption of Fog Computing in Healthcare 4.0," in *Fog Computing for Healthcare 4.0 Environments*,

S. Tanwar, Ed. Springer, Cham, 2021, pp. 1-1, doi: 10.1007/978-3-030-46197-3_1.

89. A. H. Sodhro and N. Zahid, "AI-Enabled Framework for Fog Computing Driven E-Healthcare Applications," Sensors (Basel, Switzerland), vol. 21, 2021.
90. A. Awad Abdellatif et al., "MEdge-Chain: Leveraging Edge Computing and Blockchain for Efficient Medical Data Exchange," *IEEE Internet of Things Journal*, vol. 8, pp. 15762-15775, 2021. [Online].
91. S. Manard, N. Vergos, S. Tamayo Giraldo, and F. Fontane, "Electronic health record in the era of industry 4.0: the French example," *arXiv preprint arXiv:1907.10322*, Jul. 24, 2019.
92. M. Mijwil and M. Aljanabi, "From Analog to Digitization: Rethinking Management and Operations through eHealth Integration in Industry 4.0," *Mesopotamian Journal of Artificial Intelligence in Healthcare*, pp. 27-30, 2023, doi: 10.58496/MJAIH/2023/005.
93. E. AbuKhousa, N. Mohamed, and J. Al-Jaroodi, "HEALTH 4.0: CAN HEALTHCARE INDUSTRY GET SMARTER?" in Proceedings of TMCE 2020, May 11-15, 2020, Dublin, Ireland, I. Horváth and G. Keenaghan, Eds. Organizing Committee of TMCE 2020, ISBN/EAN: 978-94-6384-131-3, p. 227.
94. S. Abbate, P. Centobelli, R. Cerchione, E. Oropallo, and E. Riccio, "Investigating Healthcare 4.0 Transition Through a Knowledge Management Perspective," *IEEE Transactions on Engineering Management*, pp. 1-14, 2022, doi: 10.1109/TEM.2022.3200889.
95. F. Syafar, M. Anwar, H. Husain, S. Suhaeb, and S. Syafar, "Puskesmas, the Third Level of Indonesia Healthcare: Current Electronic Medical Record (EMR) Status and Readiness in Meeting Industry 4.0 Challenges," in *35th IBIMA Conference*, Seville, Spain, April 2020.
96. G. P, V. Menon, L. Ramasamy, S. Kadry, and Y. Nam, "Blockchain-Based Secure Healthcare Application for Diabetic-Cardio Disease Prediction in Fog Computing," *IEEE Access*, pp. 1-1, 2021, doi: 10.1109/ACCESS.2021.3065440.
97. I. Ahmad, Dr. Abdullah, and A. Ahmed, "IoT-fog-based healthcare 4.0 system using blockchain technology," The Journal of Supercomputing, vol. 79, 2022, doi: 10.1007/s11227-022-04788-7.
98. G. Fanta, L. Pretorius, and B. Nunes, "Enabling Circular Economy in Healthcare Using Industry 4.0 Digital Technologies," pp. 1111-1123, 2021. [Online]. Available: 10.52202/060557-0085.
99. I. C. Reinhardt, J. Oliveira, and D. Ring, "Current Perspectives on the Development of Industry 4.0 in the Pharmaceutical Sector," *Journal of Industrial Information Integration*, vol. 18, p. 100131, 2020, doi: 10.1016/j.jii.2020.100131.
100. G. Daú, A. Scavarda, L. Scavarda, and V. Portugal, "The Healthcare Sustainable Supply Chain 4.0: The Circular Economy Transition Conceptual Framework with the Corporate Social Responsibility Mirror," *Sustainability*, vol. 11, no. 12, p. 3259, 2019, doi: 10.3390/su11123259.
101. N. Laurisz, M. Ćwiklicki, M. Żabiński, R. Canestrino, and P. Magliocca, "The Stakeholders' Involvement in Healthcare 4.0 Services Provision: The Perspective of Co-Creation," *Int J Environ Res Public Health*, vol. 20, no. 3, pp. 2416, Jan. 2023, doi: 10.3390/ijerph20032416.
102. D. Akanmu, N. Nordin, and U. Gunasilan, "Lean Manufacturing Practices and Integration of IR 4.0 Technologies for Sustainability in the Healthcare Manufacturing Industry," *International Journal of Service Management and Sustainability*, vol. 7, pp. 21-47, 2022, doi: 10.24191/ijsms.v7i1.17777.
103. A. Saifudin, M. Aima, A. Sutawidjaya, and S. Sugiyono, "Hospital digitalization in the era of Industry 4.0 based on GHRM and service quality,"

International Journal of Data and Network Science, pp. 107-114, 2021, doi: 10.5267/j.ijdns.2021.2.004.

104. K. Sibanda, P. Ndayizigamiye, and H. Twinomurinzi, "Industry 4.0 Technologies in Maternal Healthcare: A Systematic Review," IFAC-PapersOnLine, vol. 55, pp. 2407-2412, 2022, doi: 10.1016/j.ifacol.2022.10.069.

105. P. Arun Kumar and S. Wang, "The Design Intervention Opportunities to Reduce Procedural-Caused Healthcare Waste Under the Industry 4.0 Context – A Scoping Review," in Proceedings of the International Conference on Human Interaction and Emerging Technologies, 2021, doi: 10.1007/978-3-030-73426-8_27

106. W. Maisiri and L. van Dyk, "Leapfrogging to Industry 4.0: Lessons from the Healthcare Industry," in SAIIE29 Proceedings, 24th–26th of October 2018, Spier, Stellenbosch, South Africa, 2018, pp. 3651-1.

107. N. Akhtar, A. Haleem, and M. Javaid, "Scope of Health Care System in Rural Areas under Medical 4.0 Environment," Intelligent Pharmacy, vol. 1, 2023, Art. no. 10.1016/j.ipha.2023.07.003.

108. N. Savanović et al., "Intrusion Detection in Healthcare 4.0 Internet of Things Systems via Metaheuristics Optimized Machine Learning," Sustainability, vol. 15, no. 16, p. 12563, 2023. [Online].

109. J. Mwanza, A. Telukdarie, and T. Igusa, "Impact of industry 4.0 on healthcare systems of low- and middle-income countries: a systematic review," Health Technol (Berl), vol. 13, no. 1, pp. 35-52, Jan. 2023, doi: 10.1007/s12553-022-00714-2.

110. S. Veronika, I. Khalil, M. Atiquzzaman, and X. Yi, "Trustworthy Privacy-Preserving Hierarchical Ensemble and Federated Learning in Healthcare 4.0 With Blockchain," IEEE Transactions on Industrial Informatics, vol. 19, no. 7, pp. 7936-7945, Jul. 2023. [Online].

111. L. He, M. Eastburn, J. Smirk, and H. Zhao, "Smart Chemical Sensor and Biosensor Networks for Healthcare 4.0," Sensors (Basel), vol. 23, no. 12, p. 5754, Jun. 2023, doi: 10.3390/s23125754.

112. H. Tian, W. Ye, J. Wang, H. Quan, and C.-C. Chang, "Certificateless Public Auditing for Cloud-Based Medical Data in Healthcare Industry 4.0," *International Journal of Intelligent Systems*, vol. 2023, pp. 1-18, 2023, doi: 10.1155/2023/3375823.

113. R. Dautov, S. Distefano, and R. Buyya, "Hierarchical data fusion for Smart Healthcare," *Journal of Big Data*, vol. 6, 2019, Art. no. 6, doi: 10.1186/s40537-019-0183-6.